Java Quick Reference and Debugging Examples

Variables and types

```
int count = 3;
String name = "Ada";
boolean ready = true;
```

Use a type that matches the value; Java does not implicitly convert arbitrary strings to numbers.

Arrays and indexing

```
int[] values = {10, 20, 30};
for (int i = 0; i < values.length; i++) {
    System.out.println(values[i]);
}
```

Indexes range from 0 through length - 1. Using i <= values.length reads past the final element and throws ArrayIndexOutOfBoundsException.

Null values

```
String label = null;
if (label != null) {
    System.out.println(label.length());
}
```

Calling a method on null throws NullPointerException. Check for null or use a value that cannot be null.

Collections

```
List<String> names = new ArrayList<>();
names.add("Mina");
for (String name : names) {
    System.out.println(name);
}
```

Use the enhanced for loop when an index is not needed. Import java.util.List and java.util.ArrayList.

Methods

```
static int add(int a, int b) {
    return a + b;
}
```

A non-void method must return a value on every path. A missing return causes a compile-time error.